

Practice — Solubility of Gases

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

The dissolved gas that keeps fish and aquatic plants alive is —

- (a) nitrogen (b) oxygen (c) carbon dioxide (d) argon

Q2 • MCQ**[1]**

As temperature rises, the solubility of a gas in water generally —

- (a) increases (b) decreases (c) stays the same (d) doubles

Q3 • MCQ**[1]**

Which water holds more dissolved oxygen?

- (a) cold water (b) warm water (c) boiling water (d) all the same

Q4 • MCQ**[1]**

A mixture of a gas dissolved evenly in water is —

- (a) non-uniform (b) uniform (c) a compound (d) an element

Q5 • ASSERTION-REASON**[1]**

A: Warming a river can harm fish.

R: Warmer water holds less dissolved oxygen.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

Gas solubility _____ as the temperature of water increases.

Q7 · FILL IN THE BLANK

[1]

_____ oxygen dissolved in water keeps fish alive.

Short answer (2 marks each)

Q8

[2]

Do gases dissolve in water? Give an example of why it matters.

Q9

[2]

How does temperature affect the solubility of gases? How is this different from solids?

Q10

[2]

Why does cold water support aquatic life better than warm water?

Long / application (3 marks each)

Q11

[3]

Explain why dumping hot water from a factory into a river can harm fish.

Q12

[3]

Compare how temperature affects the solubility of a solid (like sugar) and a gas (like oxygen) in water, with one everyday example of each.

[Check yourself](#) → Answer key